

Risk Data Governance Lead

The appendices describe data I do not have. Rather than describing what I would hypothetically do, I rebuilt the landscape in Snowflake and measured it — same grains, same sign conventions, same pitfalls — then built the governance layer on top: catalog, canonical metric layer, DQ monitoring, change registries, an LLM context layer and a working agent. **Every number in this deck is a measurement.**



What is proof and what is proposal. The sandbox is a *verification method*, not the proposal. What I would put into Plata in the first 90 days is deliberately smaller than what I built to check it: four registry tables, the checks that matter on tier1–tier2, one context layer — plus an explicit list of what is **not** a priority. At \$8, checking an assertion is cheaper than debating it.

How to read this. Nine slides carry the argument. Every claim behind them is verifiable by a single query, and the queries are in the full response (`test-assignment-answer-en.pdf` , 20 pp.) together with a separate review of the source material: **20 findings across Appendices A–I, 8 of them critical**, each cited with file and line.

The governance risks are already written down — as prose

Read out of Appendices A–I and nowhere else. Every row is backed by a quote with a line number (C:7 = Appendix C, line 7).

#	Risk	DMBoK discipline	What the appendices actually say
R1	Metric with no canon	Data Quality Mgmt · Metadata	The glossary defines COR and admits "Three methodologies exist" (E:13). No formula is marked primary; the definition has no owner
R2	Sign convention in prose	Data Architecture · DQM	"Different tables use different signs... silently produces incorrect results" (C:7) — a rule across 5+ tables that exists only as a paragraph
R3	Filter column differs by schema	Data Architecture · Reference Data	dm.* filters on product_risks , tableau.* on account_type (A:42) — both accept the value 'CC'
R4	Key rule duplicated three times	Metadata · DQM	statement_num >= 2 and × 12 recorded independently in three appendices (C:99 , D:27 , F:53). Guaranteed drift at the first edit
R5 R6	Change with no trace and no version	Change Management	A table "was dropped in a recent refactor" — no date, no author, no impact list (G:47). Gen5v1 "materially shifted EL LT" (G:71): schema unchanged, nothing failed, every number now means something else
R8	SLA with no addressee	Data Governance · DQM	Expected freshness is tabulated (B:70), but not who owns the number, who is woken on a breach, or what counts as one. Ownership is not mentioned once in nine appendices
R10	PII as routine enrichment	Data Security · AI readiness	A join to ods.user_mgmt.USER_DATA is offered as an ordinary pattern: "Add demographics (gender, age, state)" (F:29) — no classification, no role limit, and an LLM agent is meant to work off these same appendices

The expensive class is not the query that fails — it is the query that runs and lies. Six of the eleven risks (R1–R4, R6, R9) return a plausible wrong number with no execution error whatsoever. The appendices warn about this twice with the word "silently", and both times the warning is addressed to the reader's memory rather than to a check.

A rule that cannot be skipped, instead of a rule that must be remembered

HOW THE DRIFT IS STOPPED

Mechanism	What it is
M1 Do not rename — translate	A mass rename across 20+ schemas breaks Tableau and never starts. Instead a name-resolution layer: old names survive but stop being a fork in the road. Zero migrations
M2 Standard applies at the entry point	Mandatory for new objects; existing ones are registered as they are, with an alias. A standard that demands rewriting the past dies in month one
M3 A convention becomes a predicate	Each rule is stored as a row with a ready SQL condition: 12 sign rules of the form <code>COR_PRNP <= 0</code> , each citing its source appendix
M4 A gate at the entrance to the layer	No object without a card (grain, filter column, sign), a named owner and steward, an SLA with an action, and a passing check run
M5 Drift is measured, not declared	A DDL snapshot is diffed against the change log. The metric is not "we introduced a standard" but "how many changes shipped unannounced"

Why a predicate and not a policy document — I tripped over this myself. The first semantic model for Cortex described the mandatory filter `statement_num >= 2` in a *comment* on the metric. A natural-language question returned **14.42%** instead of the canonical **14.62%**. Once the filter was compiled into the aggregate, the discrepancy went to zero. A comment enforces nothing — and a review policy without an automatic check is the same comment.

OWNERSHIP = THREE SEPARABLE DUTIES

Role	Accountable for
Data owner	What the number <i>means</i> ; signs off a change to the definition
Data steward	Operational quality; receives the alert when a check fails
Technical owner	The pipeline and the load SLA

The owner is whoever bears the consequences of a wrong number, not whoever wrote the pipeline. Operationally that is four obligations, each answerable by query: a signature under the definition · a named duty with a named action · a threshold in the right calendar · an obligation to give notice.

Reality test: who signed the definition, who gets woken and what they do, who had to be told in advance and how many days ahead. While any of the three is answered by a phone call, ownership exists on paper only.

"What is our actual COR for April?"

14.62%

annualised, credit cards, under the canonical definition — not because it is the nicest number, but because **the grain of the calculation matches the event the metric is defined on**.

Canon COR_PCT_CC v1.0	Value
Grain	account × statement
Model	Gen3, full total
Mandatory filter	product_risks = 'CC' AND statement_num >= 2
Denominator	SUM(avg_balance_prnp_net)
Annualisation	× 12

Provisions move when a payment is missed, and payments are attached to a statement. A daily snapshot answers a different question and mixes calendar effects into a credit metric.

THE BRIDGE: WHERE THE OTHER TWO NUMBERS COME FROM

Each step changes exactly one methodological decision and is recomputed from the data, so **the residual is zero by construction**.

Step (April 2026, product CC)	Contribution	Running
Canon — statement, Gen3, stmt ≥ 2, ×12	—	14.62%
statement_num ≥ 2 filter removed	−0.20	14.42%
grain: statement → daily	−2.14	12.28%
0-DPD bucket removed from the numerator	−0.53	11.75%
annualisation ×12 → ×365	+0.16	11.92%
= Portfolio Monitoring		11.92%
model generation Gen3 → Gen2	−0.54	11.38%
= Finance		11.38%

Two findings that reframe the task. The spread has **no constant sign**: in April the canon is the highest of the three, in March the lowest (Feb 0.74 pp · Mar 2.30 · Apr 3.24 · Jun 4.37). So "Finance always runs two points lower, just add it back" is impossible — one canon is required, not a correction factor. And the famous pitfall is third-order: the statement_num filter contributes 0.05–0.24 pp, while the choice of grain contributes 2.1–2.7. The warning is correct; the documentation's priority is not.

Three teams will drift again — the question is who notices first

The detector does not check whether a number is "right": all three are right inside their own methodology. It catches three **events**.

#	Signal	Why it is built this way
D1	The bridge residual stopped being zero	The bridge converges by construction, so a non-zero residual is the only signal that catches an unknown unknown : a calculation changed quietly, a new branch appeared, the data moved
D2	A step's contribution left its historical corridor	The grain step ranges from -7.1 to +8.1 pp across 24 months — that is normal, not broken. The threshold is built from each step's own history, or the alert fires on healthy seasonality
D3	A team's published number diverged from its registered formula	Recomputed from METRIC_DEFINITIONS and compared with what the team publishes. The only check aimed at people rather than data

Why the sign check is an aggregate, not a row check. Individual positive COR values are legitimate — that is a provision release when an account cures. A row-level rule would fire on every healthy portfolio and be switched off in week one. An alert that gets ignored is worse than no alert, and the same principle sets the D2 thresholds.

WHERE THIS SITS ON THE DQM LADDER

Rung	State in the sandbox
Basic → Defined native metrics on critical tables	12 attachments on 3 tables, daily schedule
Defined → Controlled checks on <i>meaning</i> , not form	2 custom quality functions: calendar violation and sign inversion
Controlled → Continuing traffic light on seven dimensions	22 checks: 21 green , 1 yellow

D1–D3 themselves: Defined is fully closed — the canon is recorded, all three formulas registered with a status, the bridge runs on demand. Controlled is partial: D1 and D2 compute on every rebuild, but thresholds and a schedule agreed with the owners do not exist yet.

The single yellow check is a real incident — stm_cc is 9 business days behind against an SLA of 5. Tellingly, I first put that delay only into the ETL log while the data itself was current, and the traffic light showed all green. The monitor was measuring something other than what it claimed. That is exactly the class of error that makes a DQ dashboard decorative.

Gen5v1 re-estimated: a data change with no schema change

Nothing fails, no query errors — every dashboard quietly starts showing different numbers, which makes it **more dangerous than dropping a table**. A drop is noticed in a minute; a 35% shift in EL LT for 2025 H1 vintages is noticed a quarter later, in a variance review.

THREE DECISIONS TO CLOSE BEFORE DOING ANYTHING

Decision	Who decides, and why them
Retrospective closed periods or forward only	Head of IRM with Finance — monthly reporting breaks on exactly this. The consequence lands on reporting, not on modelling
Form of publication new snapshot or overwrite	The mart owner. Here: a new 2026-06 snapshot, 2026-03 left alive. That is versioning, not "a backup just in case"
Life of the old base until when may a consumer stay	The mart owner with the consumer. Without a date the pin is permanent and the organisation keeps two truths forever

THEN, IN ORDER

- **Measure the shift before announcing it** — ×1.35 on EL LT for 2025-01...2025-06 vintages. "The model was re-estimated" without a number is useless.
- **Notify from the registry**, version the mart, log the change with a breaking flag and the fact of notice.
- **Keep a rollback path**: switching a view, not reloading data — both snapshots stay live, so rollback costs seconds. A plan that needs a restore from backup is never used.
- **Close on evidence, not on the date**: retire the old view once access history shows nobody is left on it. While someone still reads the previous version, truth has multiplied rather than migrated.

IMPACT ASSESSMENT: A QUERY, NOT A PHONE ROUND

Consumer	Notice	What actually breaks it
CLIP decision model	30 days	The EL LT shift — the model is calibrated on it
Monthly reporting (CUBE)	20 days	Value changes on closed periods; a new snapshot is fine
Statement COR dashboard	10 days	Any change to forecast values or grain
LLM agent	0 days	Column rename or drop. A change in <i>values</i> needs a context refresh, not a notice

The agent row matters: a human needs warning in order to rebuild a report; an agent gains nothing from warning — it needs the context layer to update **at the same moment as the data**.

CM ladder	Implemented — from Basic, where you find out afterwards
Defined	CHANGE_LOG , 4 records, 3 breaking
Controlled	impact view, versioning, parallel window, a tombstone instead of "object does not exist"
Continuing	DDL snapshots and hash reconciliation: a changed hash with no log entry is a verdict, not an opinion

The consumer that is not in the registry: the agent's golden question set depends on the model itself. After the next re-estimation its expected values go stale, and a regression drop becomes indistinguishable from agent degradation. Found, openly not yet closed.

The appendices are good documentation for a human

An agent lacks three things: **structure instead of prose** (a row it can find by object name, not a sentence) · **term resolution** (the glossary says TDC is a credit card, not that in a query it is `product_risks = 'CC'`) · **rules inside the data, not beside it** (a mandatory filter in a document can be ignored; compiled into the metric it cannot).

BUILT — AND THEN ACTUALLY USED BY AN AGENT

Object	n	What it gives the agent
Table cards	9	Grain, filter column, sign, mandatory filters, a "do not" section
Join graph	10	The exact expression and the trap on each edge
Pitfalls	10	Symptom, rule, executable detection query
Glossary	17	Term → synonyms → object/column/value
Metric contracts	4	The only sanctioned way to compute
Golden question set	16	Regression test, expected values from the data

The whole layer is **13,664 characters ≈ 3.4k tokens** — it fits in the system prompt entire, so "will the documentation even fit" never arises.

The agent in operation, not on paper. RISK_COPILOT inside Snowflake, three tools: the semantic view `SV_COR`, Cortex Search over Appendices A–I (**86 chunks**, documents never leave the account), and read-only SQL under `LLM_AGENT_R0`. Surfaced through Telegram behind an allowlist. A dbt project runs from git (**PASS=11, ERROR=0**) whose test fails the build if the transformation layer diverges from the canon by more than 0.01 pp; a daily alert watches for drift.

SCORED AGAINST A PLAN WRITTEN BEFOREHAND — 23 QUESTIONS

Verdict	n	What it shows
PASS	11	The governance objects hold up under an agent: the dropped table redirects , "give me demographics of delinquent customers" is refused rather than routed around, April COR comes back at 14.62% <i>with the contract named</i>
FAIL	2	Business days in April: 22 against 20 — no grant on the holiday calendar, so the agent answered from arithmetic, plausibly and wrongly, with no error anywhere. Closed with a point grant
PARTIAL	10	Consistently: right table, right derivation, right trap — but could not execute . The limit is in the grants, not the model, and it surfaced only because the run was scored

The score is not the deliverable — the existence of a score is. Without a plan and a golden set, "the agent works" is a demo.

TRUST IS A PRODUCT, NOT A SUM

privileges × measured correctness at a threshold × answer provenance × the ability to review what it did. **Zero any factor and the result is zero**, not "partially trusted".

Gap	Status
G1 Pass threshold	closed — baseline captured; -- compare fails the run on degradation, by difficulty
G2 Resource limits	closed at the service boundary — 90-second sessions on a dedicated warehouse, against a role default of 48 hours
G3 Answer provenance	partial — the contract is named; model snapshot and data date are not
G4 Trust is binary	open — analyst / reporting / regulatory need three access classes. First thing I would close in the role

Two dimensions first — and an explicit list of what is not a priority

WHERE PLATA STANDS, ON THE EVIDENCE

Discipline	Level	Confidence
Data Quality Mgmt	Basic	High — freshness is tabulated, but no threshold, owner or trace of a check anywhere
Change Management	Basic	High — a drop described as a fact, a re-estimation as a heads-up
Data Catalog	Defined by content Basic by ownership	Thin — a composite score on purpose: averaging it would hide the very attribute that is missing
Reference Data	Defined	Thin and indirect — only product codes are visible

Working definition used throughout: Basic — the practice lives in people's heads · Defined — the rule is written and has an owner · Controlled — compliance is checked automatically and breaches escalate to a name · Continuing — the control system itself is measured.

Not a 90-day priority, deliberately. Reference Data Management (an alias reference closes 90% of the pain in a day) · Master Data Management for customers · replatforming or buying a catalog (a tool bought before the process works preserves the absence of process) · ownership coverage of tier3 (tagging forty technical tables creates the appearance of governance).

DQM AND CHANGE MANAGEMENT, BASIC → DEFINED

Not only because the assignment names them the sharpest pain, but for the dependency order: **without DQ you cannot prove a change broke something, and without change management any quality gain is rolled back by the next release.** In parallel — the canonical metric layer: you cannot monitor the quality of a metric whose definition is not fixed.

Gate	What is done	How it is evidenced
Day 30	Five measurements taken; the metric canon fixed and signed; the bridge between the three calculations computed	The decomposition reproduces from data with a zero residual. Priorities explicitly confirmed or revised
Day 60	Checks on every tier1 table with threshold, steward and prescribed action; statement freshness in business days; change log established	No Monday false alarms; every firing signal has an addressee
Day 90	Consumer registry with notice periods; schema snapshot reconciliation running automatically	No breaking change without a record and a notice; reconciliation finds nothing unlogged

What I measure in the first 30 days — two of the five can overturn this plan, and that is the point: how many versions of one metric are actually in circulation · share of incidents found by monitoring before the user (the only one that *cannot* be measured retroactively, so the incident log with a mandatory "who found it" field is set up in week one) · share of schema changes that were registered · time to answer a typical analyst question · agent pass rate on the golden set.

The main risk of the plan is not technical. For half of the tier1 objects the technical owner is **DWH Engineering** — a team that does not report to this role. The workaround is built into the design: every check is constructed **on the risk side** and requires no change to upstream pipelines. Control is placed where the authority is, not where the source of the problem sits — otherwise the plan depends on someone else's quarterly roadmap.

Two Data Stewards: remove work first, add work second

The two analytics engineers are not idle capacity — a plan that adds work to a saturated queue is not a plan. **Managing two people is a capacity decision before it is a people decision.**

FIRST TWO WEEKS: MEASURE THEIR TIME, NOT THE DATA

What I measure	What decision it changes
Structure of incoming work — ad-hoc / extract / incident / development	If more than half is ad-hoc, the context layer moves <i>ahead</i> of DQ checks: it removes work rather than adding it
Repeat rate of questions	A repeated question is a documentation defect, not an analyst request. Fixed once, in the glossary or a card
Interrupt load	Two people cannot absorb both an ad-hoc queue and a delivery plan
Single points of knowledge	Bus factor is a governance risk, not an HR one. It sets the rotation
Where the queue actually stalls	Separates "we are slow" from "we are blocked by DWH" — only one of them is mine to fix

The appendices predict most of the toil: three COR versions mean somebody reconciles by hand before every management meeting; "silently produces incorrect results" means every query is re-verified by attention rather than by a check; broken cross-references mean the same questions answered in chat again and again; weekend-stale means a false alarm triaged every Monday. **Every one is toil manufactured by the absence of governance** — hence the sequencing rule: the first delivery must give them hours back, or the second never gets built.

SPLIT OWNERSHIP, NOT TICKETS

	Steward 1 — marts and quality	Steward 2 — meaning and registries
Owens	Checks on the marts, freshness in business days, the DQ scorecard, incident duty	Metric definitions, glossary, product reference, table cards, change log, consumer registry
Day 90	tier1 covered; zero Monday false alarms	Canon signed; context layer extended to the full risk layer
Grows into	DQ engineering: predicates, dbt tests, drift invariants	Semantic and metadata layer: contracts, agent readiness

Both are named in `META.OWNERSHIP` as stewards of their own objects, with the action they take when a check fails. **The ownership model this response proposes for data is the model I run the team on.** Duty rotates weekly; both present at a gate at least once.

HOW THE EXPERIENCE TRANSFERS

- **Pair on the first, solo on the rest** — the first sign predicate together, the other eleven theirs. The teaching cost is paid once per *class* of work, not per task.
- **They author the standard, I review it** — one written by the lead gets complied with; one written by the steward gets defended. The naming check missing from the 26 is the first thing I hand over rather than close myself.
- **"Explain the number" duty** — whoever owns a definition answers the Head of Risk directly, with me in the room and not in front.
- **Review with the reason attached** — every rejection names the check that would have caught it. Review that only says no teaches obedience.

How I know the delegation is real: share of checks authored by the stewards rather than by me — near zero at day 30, above two thirds at day 90. Not delegated: the signature under the canon, the right to refuse at the gate, the agent's trust threshold. Never kept: writing all the checks myself — a lead who builds everything personally ends up with two executors and a bus factor of one.